Exercises 5
(Functions- Recursion)

1.1 What is the output of the following code?

a)
```
void puzzle1(int n)
{
     if (n == 0)
        return;
     cout << n << " ";
     puzzle1(n - 1);
}
void main()
{
     puzzle1(3);
}
```

b)
```
void puzzle2(int n)
{
     if (n == 0)
          return;
     puzzle2(n - 1);
     cout << n << " ";
}
void main()
{
     puzzle2(4);
}
```

c)
```
void puzzle3(int n)
{
     if (n == 0) return;
     cout << n;
     puzzle1(n - 1);
     cout << n;
}
void main()
{
     puzzle3(3);
}
```

d)
```
int puzzle4(int n)
{
     if (n == 1)
        return 1;
     return n + puzzle4(n-1);
}
void main()
{
     cout<<puzzle4(4);
}
```

1.2 Create a **recursive** function **SumDigits** that takes a number as input and returns the sum of its digits. For example, if the input number is 123, should return **6**.

1.3 Write a **recursive** function to multiply two numbers **a** and **b** using only addition (+). For example, multiply(3, 4) → 3 + 3 + 3 + 3 = 12.

1.4 Write a **recursive** function that reverses a string. For example, reverse("hello") should return "olleh".

1.5 Write a **recursive** function to find the largest element in an array of integers. For example, **findMax**([3, 1, 8, 4]) should return 8.

1.6 Write a **recursive function** to search for a specific value within an array. The function should return the **index** of the value if found, or **-1** if it does not exist**.**

1.7 Create a **recursive** function named **IsPalindrome** that takes a string parameter and returns a boolean value indicating whether the string is a palindrome For example: "RADAR" is a palindrome.